

Math 6 — Q1 Mid-Quarter Test · ANSWER KEY

Mid-Q1 Content · 15 questions

Teacher key. Student answers may use equivalent forms (fractions, decimals, with or without units).

1. Convert 5.2 km to meters.

Answer: 5,200 m

2. Convert 250 g to kilograms.

Answer: 0.25 kg

3. Convert 3.5 L to milliliters.

Answer: 3,500 mL

4. Convert 1500 mm to centimeters.

Answer: 150 cm

5. Convert 0.075 kg to grams.

Answer: 75 g

6. The actual mass of an object is 50 g. A student measured 48 g. What is the percent error?

Answer: 4%

7. Actual density of water is 1.00 g/mL. A student measured 0.95 g/mL. Percent error?

Answer: 5%

8. A blue whale weighs 180,000 kg (actual). A scientist measured it as 175,000 kg — an error of 5,000 kg. A house cat weighs 8 kg (actual). Another scientist measured it as 7 kg — an error of 1 kg. Which measurement is MORE accurate?

Answer: whale (whale $\approx$ 2.78% error vs. cat = 12.5% error)

Why? Percent error compares the mistake to the size of what's being measured. The whale error of 5,000 kg sounds huge, but relative to 180,000 kg it's less than 3%. The cat error of 1 kg is much smaller in absolute terms but 12.5% of 8 kg. Smaller percent error = more accurate.

9. Evaluate: $-8 + 5$.

Answer: -3

10. Evaluate: $-7 - (-3)$.

Answer: -4

11. Evaluate: $(-6) \times 4$.

Answer: -24

12. Evaluate: $(-36) \div (-9)$.

Answer: 4

13. Simplify: $-(-12)$.

Answer: 12

14. Which is larger: -5 or -8 ?

Answer: -5

15. What is $|-15|$ (absolute value of -15)?

Answer: 15